SonoSPADE: Supplementary Material

Anonymous

Anonymous Institution

This supplement collects the whole-frame FID/KID scores (Table S1), the training-term ablation (Table S2), the per-tissue label-swap locality breakdown (Table S3), and the per-tissue qualitative result (Fig. S1) referenced in the main paper. All numbers are generated from the same result files as the main tables.

Table S1. Whole-frame FID/KID ($\times 10^3$) against the real pool (InceptionV3), lower better. SonoSPADE is not competitive by design, because these metrics reward the global recolor it does not perform. The bottom block re-runs in the sector geometry, which improves all absolute scores but not the ranking.

Method	FID ↓	KID $\times 10^3$ ↓
Raw physics	315.3	294.3
CycleGAN	**192.2**	203.1
CUT	235.6	245.9
S-CycleGAN (reimpl.)	199.7	**196.9**
SonoSPADE	294.7	306.5
Raw physics-curvi	212.9	265.4
CycleGAN-curvi	**67.9**	**55.6**
CUT-curvi	72.3	55.9
S-CycleGAN-curvi	104.2	84.2
SonoSPADE-curvi	191.1	232.6

Table S2. Ablation (one term off per row, each retrained at a fixed budget). Removing the frozen S_{US} seg-consistency term roughly halves locality and degrades structure, the largest single effect. This is the full version of the ablation summarized in Sec. 4.5 of the main paper.

Variant	Edge IoU ↑	Organ W_1 ↓	Locality ↑
Full SonoSPADE	0.41	0.115	27.1
− S_{US} consistency	0.30	0.129	13.7
− OASIS LabelMix	0.39	0.111	28.6
− temporal consistency	0.42	0.106	26.3

Table S3. Per-tissue label-swap locality on the physics/label content frames (higher means the texture change is more localized to the relabeled region). Liver is the reference tissue and is excluded; spleen and muscle rarely reach the 200-pixel minimum in the content frames and are omitted. The values are the measured per-class means behind the main-paper locality of 26.3 (they average to it). Per-frame standard deviations can be added with `scripts/eval_locality_std.py`.

Tissue	Label-swap locality ↑
Gallbladder	22.8
Vessel	18.7
Kidney	30.2
GI tract	40.3
Pancreas	28.8
Fat	10.4
Bone	60.7
Lung	13.0
Soft tissue	11.5
All tissues (mean)	26.3
Label-free translators	0 (by construction)

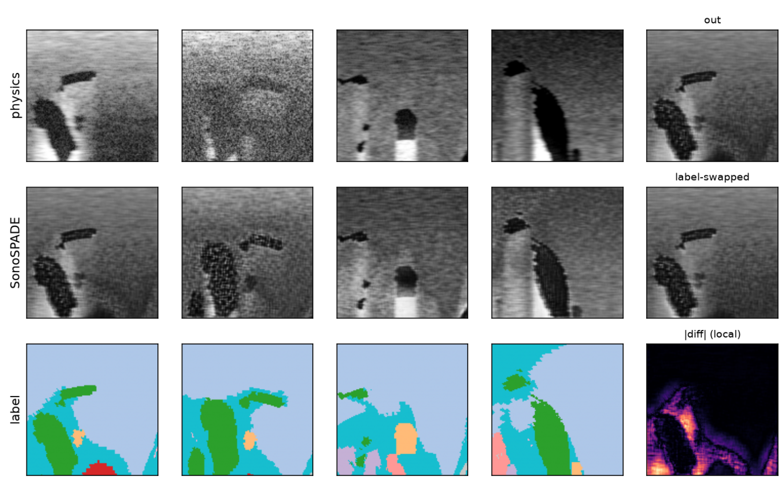

Fig. S1. Per-tissue qualitative result (Sec. 4.2 of the main paper). The rows are the physics render, the SonoSPADE output, and the tissue label. Each tissue gets its own speckle, and relabeling a region changes its texture locally (last column): the swapped region's texture changes while the rest of the frame stays put, which is what the label-swap locality metric measures.